

PLIKD: Prompt Learning with Instance-aware Knowledge Distillation for Web-scale Semantic Image Classification

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Abstract

With the rapid growth of multi-modal content on the Web, robust vision-language models are essential for semantic understanding and classification of web images under diverse and dynamic contexts, supporting Web applications such as multimedia search and recommendation. Prompt learning has proven effective for enhancing vision-language models in semantic image classification tasks. However, previous methods often suffer from poor generalization: the learned prompts tend to overfit the base classes seen during training, leading to poor performance on unseen classes and under distribution shifts. This issue is especially challenging in Web-scale data, where new classes emerge and distributions shift dynamically.

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To address these limitations, we propose PLIKD, a novel prompt learning method that integrates instance-aware knowledge distillation for robust Web-scale semantic image classification. Specifically, PLIKD introduces an instance-aware knowledge extraction module, which leverages multi-modal large language models through a step-by-step strategy to extract external knowledge for each image instance. To incorporate this extracted knowledge, PLIKD further introduces an instance-aware knowledge distillation module, which consists of two key steps: (1) a dual-teacher strategy for robust and informative knowledge distillation, and (2) fine-grained cross-modal alignment via Smooth and Sparse Optimal Transport. Extensive experiments demonstrate that PLIKD significantly improves generalization to both seen and unseen classes, and remains robust under distribution shifts, outperforming existing state-of-the-art methods on Web-scale semantic image classification.

CCS Concepts

• Information systems → Web applications; • Computing methodologies → Artificial intelligence.

Keywords

Web-scale Semantic Image Classification, Knowledge Distillation, Prompt Learning, Multi-Modal Large Language Models

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1 Introduction

The Web has become the largest source of multi-modal content, with massive collections of images, text, and user-generated data. Understanding such content is crucial for Web applications such as multimedia search [23, 68], recommendation [26], and online content moderation [11, 19]. Unlike static datasets, Web data are large-scale and continuously evolving, where new classes emerge and distributions shift dynamically. To tackle these challenges, robust vision-language models (VLMs) are essential for semantic understanding and classification of web images under diverse and dynamic contexts. Prompt learning methods have shown promising results in adapting VLMs to image classification tasks [66, 67]. These methods initialize prompts as a set of learnable vectors and optimize them by minimizing the distance between visual features and prompt features. This enables the model to adapt to specific tasks without extensive retraining. Nevertheless, in these methods, the learned prompts remain static and cannot adapt to new classes, significantly limiting their generalization on Web-scale data, where new classes continuously emerge and distributions shift dynamically. Thus, how to enhance model performance on unseen classes and under distribution shifts has become an urgent problem.

In this paper, we focus on the problem of Web-scale semantic image classification under unseen classes and distribution shifts. That is, some target samples may belong to classes that are not present in the base training set, and the data distribution may vary significantly across datasets. Such settings are common in real-world scenarios, particularly on the Web, and are substantially more challenging than conventional image classification tasks, which typically assume a closed set of classes and a consistent data distribution. Specifically, there are two main challenges: **CH1**: How can we improve the model's ability to classify target samples that belong to unseen classes? **CH2**: How can we ensure robust performance on known classes under distribution shifts?

However, existing methods cannot solve **CH1** and **CH2** well. On the one hand, the learned prompts in these methods tend to overfit the base classes observed during training, as they are optimized to minimize the loss on a fixed set of training classes. Consequently, when applied to unseen classes, these prompts fail to generalize effectively, even within the same dataset. Although CoCoOp [66] introduces conditional prompt learning to improve generalization to unseen classes, it relies on a lightweight network with limited capacity for extracting instance-specific information. This limitation leads to insufficient instance representations and makes the model's performance on unseen classes lag behind CLIP in many

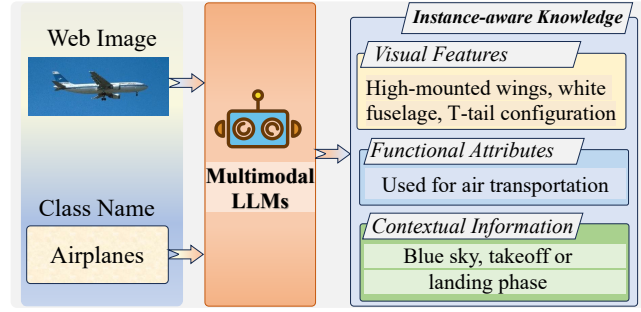


Figure 1: Illustration of the composition of instance-aware knowledge in our method.

cases. Therefore, traditional models cannot solve **CH1** well. On the other hand, existing approaches neglect the potential of prompts as effective carriers of broader knowledge [3, 14]. The ability of prompt to incorporate and propagate external knowledge is not fully leveraged. Specifically, in these methods, the learned prompts remain static and cannot flexibly incorporate external knowledge [66], significantly limiting their generalization when data distributions shift. Thus existing methods cannot solve **CH2** well. As a result, existing methods demonstrate suboptimal performance on both in-distribution (ID) and out-of-distribution (OOD) tasks [20]. In conclusion, previous methods cannot provide satisfactory results in tackling the Web-scale semantic image classification.

To address these limitations, we propose Prompt Learning with Instance-aware Knowledge Distillation (PLIKD), a novel method for image classification under unseen classes and distribution shifts. To enable our method to dynamically adapt to each instance and significantly improve robustness in challenging conditions (solving **CH1** and **CH2**), we utilize two main components in PLIKD, i.e., *instance-aware knowledge extraction module* and *instance-aware knowledge distillation module*. The instance-aware knowledge extraction module aims to extract external knowledge for each input image, which we term instance-aware knowledge. We define instance-aware knowledge as a multidimensional representation comprising visual features, functional attributes, and contextual information, offering a comprehensive description of instances. Fig. 1 illustrates the composition of this knowledge in our method. We leverage a Multi-Modal Large Language Model (MLLM) as the knowledge extractor and employ a step-by-step prompting strategy to dynamically generate the instance-aware knowledge for each instance. The instance-aware knowledge distillation module is set to transform the extracted knowledge into a distilled representation tailored to each image via a dual-teacher strategy. This distilled representation is then concatenated with the learnable prompt to form an **instance-aware knowledge prompt**. We then employ Smooth and Sparse Optimal Transport (SSOT) [7] to perform fine-grained cross-modality alignment [51, 69] between the instance-aware knowledge prompt features and the corresponding image features. By incorporating these two modules, PLIKD can effectively utilize instance-aware knowledge to both enhance the model's ability to recognize instances from unseen classes (**CH1**) and improve generalization under distribution shifts (**CH2**).

We summarize our main contributions as follows: (1) We propose a novel method, PLIKD, for Web-scale semantic image classification under unseen classes and distribution shifts. PLIKD integrates

instance-aware knowledge distillation with prompt learning for robust generalization. (2) We leverage an MLLM as a knowledge extractor with a step-by-step prompting strategy to dynamically generate instance-aware knowledge for each instance. (3) We propose a dual-teacher strategy to distill the extracted knowledge into a robust and semantically aligned representation. This distilled representation is then concatenated with learnable prompts to form instance-aware knowledge prompts. We further adopt SSOT for fine-grained cross-modality alignment. (4) Extensive experiments demonstrate that PLIKD significantly improves generalization on both seen and unseen classes, and remains robust under distribution shifts, outperforming existing state-of-the-art methods.

2 Preliminaries

Introduction to Optimal Transport. We provide a brief introduction to Optimal Transport (OT) [15] to facilitate a better understanding of the paper. For two random variables $X = \{x_1, \dots, x_i, \dots, x_n\}$ and $Y = \{y_1, \dots, y_j, \dots, y_t\}$, we denote their corresponding probability measures as r and c . The cost matrix denoted as M , measures the cost of transferring from X to Y . Based on the above description, we can give the OT representation of a discrete distribution:

$$r = \sum_{i=1}^N p_i(x_i) \delta(x_i), \quad c = \sum_{j=1}^T p_j(y_j) \delta(y_j), \quad (1)$$

where $p_i(x_i)$ and $p_j(y_j)$ represent the probability masses of x_i and y_j respectively and their sum is 1, r and c are set as uniform distributions: $r_i = \frac{1}{N}$ for all i and $c_j = \frac{1}{T}$ for all j . N and T are the number of variables in X and Y , respectively. Then OT distance can be expressed as:

$$d_M(r, c) = \min_{P \in U(r, c)} \langle P, M \rangle, \quad (2)$$

where $U(r, c) := \{P \in \mathbb{R}_{\geq 0}^{N \times T} \mid P1_T = r, P^T 1_N = c\}$. We can get the transportation matrix P by minimizing the distance $d_M(r, c)$.

The solutions to classic OT problems suffer from excessive computational costs, especially for large-scale data. In order to tackle this issue, PLOT [4] uses entropic OT [7] to optimize with lower time consumption. The new formulation with a Lagrange multiplier of the entropic regularization can be defined as follows:

$$d_M^\lambda(r, c) = \langle P^\lambda, M \rangle, \quad (3)$$

where

$$P^\lambda = \arg \min_{P \in U(r, c)} \langle P, M \rangle - \lambda h(P), \quad (4)$$

where $\lambda \in [0, +\infty]$ and $h(P) = -\sum_{i=1}^N \sum_{j=1}^T P_{ij} \log P_{ij}$. However, the addition of the entropic regularization term inevitably introduces bias in the OT matrix, leading to the matching of unrelated variables in X and Y . To address this issue, we adopt SSOT [1], which encourages sparse solutions, to solve the original OT problem:

$$d_M^\epsilon(r, c) = \langle P^\epsilon, M \rangle, \quad (5)$$

where

$$P^\epsilon = \arg \min_{P \in U(r, c)} \langle P, M \rangle + \frac{\epsilon}{2} \|P\|_2^2, \quad (6)$$

where ϵ is the coefficient that controls the sparsity of the transport matrix. Then, we can obtain a more accurate P^ϵ by formulating the dual form and solving SSOT using an alternating optimization approach. We introduce the solution procedure of SSOT below.

Algorithm 1 The optimization procedure on SSOT.

Require: Initialize cost matrix C , sparsity coefficient ϵ , probability vectors μ and ν .

- 1: **repeat**
 - 2: Solve f in Eq. (11).
 - 3: Solve g in Eq. (12).
 - 4: Calculate the matching matrix in Eq. (9).
 - 5: **until** Converge
 - 6: **return** The transport matrix π .
-

Solution procedure of SSOT. We provide an explanation of how alternating optimizations is used to solve SSOT. First, we revisit the definition of SSOT: given two probability vectors $\mu \in \mathbb{R}^{m \times 1}$ and $\nu \in \mathbb{R}^{n \times 1}$, and a cost matrix $C \in \mathbb{R}^{n \times m}$, the objective that we aim to optimize is:

$$\begin{aligned} \min_{\pi} l &= \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} C_{ij} + \frac{\epsilon}{2} \sum_{i=1}^n \sum_{j=1}^m (\pi_{ij})^2 \\ \text{s.t. } &\sum_{j=1}^m \pi_{ij} = \nu_i, \sum_{i=1}^n \pi_{ij} = \mu_j, \pi_{ij} \geq 0, \end{aligned} \quad (7)$$

where $\pi \in \mathbb{R}^{n \times m}$ is the transport plan, ϵ represents the sparsity regularization coefficient. By introducing two multipliers f and g , Eq. (7) can be transformed into an unconstrained dual form:

$$\begin{aligned} \max_{f, g} \min_{\pi} l &= \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} C_{ij} + \frac{\epsilon}{2} \sum_{i=1}^n \sum_{j=1}^m (\pi_{ij})^2 \\ &+ \sum_{i=1}^n f_i \left(\nu_i - \sum_{j=1}^m \pi_{ij} \right) + \sum_{j=1}^m g_j \left(\mu_j - \sum_{i=1}^n \pi_{ij} \right). \end{aligned} \quad (8)$$

Taking the differentiation w.r.t on π we can obtain that:

$$\pi_{ij} = \max \left(\frac{f_i + g_j - C_{ij}}{\epsilon}, 0 \right). \quad (9)$$

By substituting Eq. (9) into the constraint of Eq. (7), we obtain:

$$\sum_{j=1}^m \left[\frac{f_i + g_j - C_{ij}}{\epsilon} \right]_+ = \nu_i, \quad (10)$$

where $\left[\frac{g_j + f_i - C_{ij}}{\epsilon} \right]_+$ denotes the positive part of $\frac{g_j + f_i - C_{ij}}{\epsilon}$. By simplifying Eq. (10), we can obtain:

$$\sum_{j=1}^m [f_i - (C_{ij} - g_j)]_+ = \epsilon \nu_i. \quad (11)$$

Likewise, we can derive an equation with respect to μ_j :

$$\sum_{i=1}^n [g_j - (C_{ij} - f_i)]_+ = \epsilon \mu_j. \quad (12)$$

We can solve it with Algorithm 1 efficiently.

3 Modeling for PLIKD

In this section, we will introduce the details of our proposed method PLIKD. First, we describe the notations and problem definition. Suppose we have a pretrained CLIP model A consisting of a visual encoder $V(\cdot)$ and a text encoder $T(\cdot)$, and a dataset $D \subset X \times Y$, where each sample is an image x with its corresponding label y .

Our goal is to adapt the CLIP model to the dataset D in a few-shot learning setting. Specifically, we aim to fine-tune the model using only a small number of labeled samples (e.g., 16 samples per class) and improve its performance on the test set of D . This task is commonly referred to as few-shot image classification, where the model must generalize to seen and unseen classes with limited training data. In addition to class generalization, the model should maintain robustness under distribution shifts, where the test samples may come from a different distribution than the training set.

The framework of PLIKD is illustrated in Fig. 2. PLIKD includes two primary components, i.e., *instance-aware knowledge extraction module* and *instance-aware knowledge distillation module*. The instance-aware knowledge extraction module is set to extract multidimensional instance-aware knowledge for each image in the training set. The instance-aware knowledge distillation module aims to transform the extracted knowledge into a distilled representation tailored to each image. Combining these two modules, PLIKD can effectively integrate instance-aware knowledge to enhance the model's ability to adapt to diverse instances and improve its robustness to both unseen classes and distribution shifts. We will introduce the model details in the following subsections.

3.1 Instance-aware Knowledge Extraction Module

In this section, we will first introduce the details of the instance-aware knowledge extraction module. To address the limitations of existing methods that are optimized solely for class-level information, we integrate an instance-aware knowledge extraction module into PLIKD. This module employs an MLLM as the knowledge extractor and adopts a step-by-step prompting strategy to extract multidimensional instance-aware knowledge.

Instance-aware Knowledge. We decompose instance-aware knowledge into a multidimensional representation comprising visual features, functional attributes, and contextual information, capturing a more comprehensive representation of each instance. Fig. 1 illustrates the composition of instance-aware knowledge in our method PLIKD. Visual features refer to the distinctive visual characteristics of an instance. Functional attributes describe the role or function of an instance. As demonstrated in CoOp [67], task-relevant context can significantly improve performance. For instance, adding “a type of flower” for Flowers102 [39], “texture” for DTD [6], and “centered satellite photo” for EuroSAT [16] enhance the model's understanding. This insight motivates the inclusion of functional attributes, as they offer supplementary information that complements visual features. Contextual information refers to the environment or setting in which an instance typically appears. For the class “airplanes”, contextual information may include “airports” or “blue sky”. Contextual information is particularly important when visual features alone are insufficient or ambiguous.

Knowledge Extraction from MLLMs. To guide the MLLMs in generating higher-quality and more informative features, we employ CoT prompting [47, 54, 55, 61]. This structured approach enables the model to extract multidimensional instance-aware knowledge in a step-by-step manner, covering visual features, functional attributes, and contextual information. In summary, we design a three-step prompting strategy as follows:

Step 1. To establish a foundational understanding of the instance, we first provide the class name (e.g., “panda”) to the MLLMs. The model then generates initial instance-aware knowledge q across three key dimensions: visual features (e.g., *black and white fur, plump body*), functional attributes (e.g., *herbivorous animals*), and contextual information (e.g., *bamboo forests, zoos*). This step enables the model to retrieve general knowledge about the class, forming a basis for further refinement. We prompt the MLLMs with the following question, referred to as Question 1:

Question 1: *Based on the given class name, describe its typical visual features, functional attributes, and contextual information. Be concise and ensure accuracy.*

This step can be formulated as follows:

$$q = \text{MLLMs}(\text{Question1, class name}). \quad (13)$$

To ensure that the MLLMs generate outputs aligned with our requirements, we incorporate explicit illustrative examples within Question 1 to demonstrate the desired response format and content. A detailed introduction and analysis can be found in the Appendix. The following briefly illustrates a representative example employed in Question 1:

Example: *Class: airplanes. Visual Features: Wings, fuselage, engines, tail. Functional Attributes: Used for air transportation. Contextual Information: Airports, blue sky.*

Step 2. Next, we incorporate the image x alongside the initial instance-aware knowledge q from **Step 1** to refine the extracted knowledge. The MLLMs supplement missing features and remove irrelevant ones based on the specific instance. By directly analyzing the image, the model can identify or correct certain attributes that might have been overlooked when relying solely on the class name. This refinement ensures a more accurate and instance-specific representation. We guide the model in generating the refined knowledge q_r based on the following template question:

Question 2: *Given the provided image and instance-aware knowledge, refine the knowledge by supplementing missing but relevant features and removing unrelated information. Ensure that the descriptions remain concise and aligned with the specific image.*

This step can be formulated as follows:

$$q_r = \text{MLLMs}(\text{Question2, } x, q). \quad (14)$$

Step 3. Finally, we combine the refined instance-aware knowledge q_r from **Step 2**, the image x , and the class name to obtain detailed and precise multi-modal information about the instance. This step enables the MLLMs to extract fine-grained visual attributes (e.g., *skin color, texture*) and other instance-specific characteristics that were not captured in the previous steps. By integrating information across different modalities, the model ensures a more accurate and comprehensive understanding of the instance. We guide the model in generating the multi-modal knowledge q_m based on the following template question:

Question 3: *Please combine the refined instance-aware knowledge, the image, and the class name to obtain detailed and precise information about the instance. Focus on specific instance attributes such as visual details (e.g., skin color, texture) or contextual nuances. Ensure accuracy and keep the response concise.*

This step can be formulated as follows:

$$q_m = \text{MLLMs}(\text{Question3, } q_r, x, \text{class name}), \quad (15)$$

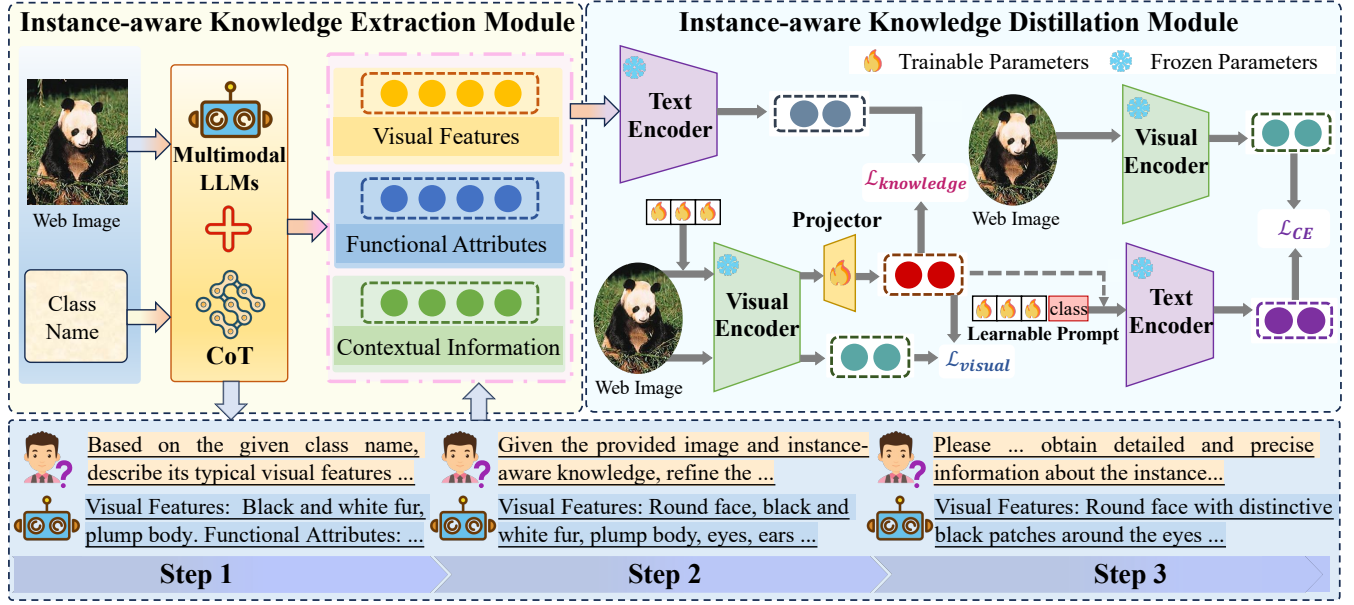


Figure 2: The model framework of our proposed PLIKD model.

$$q_m = (q_v, q_f, q_c), \quad (16)$$

where q_m includes three components: visual features q_v , functional attributes q_f , and contextual information q_c .

3.2 Instance-aware Knowledge Distillation Module

After we obtain the instance-aware knowledge for each image from the instance-aware knowledge extraction module, we should leverage this external knowledge to enhance image classification. Since MLLMs may introduce hallucination and incur high computational costs during inference, it is crucial to inherit the useful knowledge from MLLMs while mitigating these issues. To fulfill this task, we propose the instance-aware knowledge distillation module to distill the knowledge from MLLMs into PLIKD for efficient and accurate image classification. This module includes two main steps, i.e., dual-supervised distillation and cross-modal alignment. Dual-supervised distillation aims to learn a distilled representation that encodes external knowledge while retaining essential visual semantics. Cross-modal alignment aims to achieve fine-grained and precise alignment between the prompt features and visual features. By combining these two steps, our model can efficiently leverage rich instance-aware knowledge to enhance the performance.

Dual-supervised Distillation. First, we employ the text encoder $T(\cdot)$ to extract the textual features of the instance-aware knowledge as follows:

$$t_f = T(q_m). \quad (17)$$

Meanwhile, we employ the visual encoder $V(\cdot)$, and combine it with a learnable visual prompt l_v and a learnable feature projector $g(\cdot)$ (see Fig. 2) to obtain the instance representation z_r and visual features F_x for the input image x :

$$z_r = g(V(x; l_v)), \quad F_x = V(x). \quad (18)$$

Then we should consider the matching relationship between the instance representation z_r and instance-aware knowledge features t_f . Therefore we propose a dual-teacher strategy to transform the instance-aware knowledge extracted from MLLMs into z_r . Specifically, we set MLLMs as the first teacher and encourage z_r to align with t_f via a distillation loss as follows:

$$\mathcal{L}_{knowledge} = \frac{1}{N} \sum_{i=1}^N \left[-\log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^N \exp(s_{ij}/\tau)} \right], \quad (19)$$

where τ is a temperature parameter, s_{ij} denotes the cosine similarity between the z_r^i and the instance-aware knowledge feature t_f^j , and N denotes the number of samples in a batch. To prevent the learned representation from deviating too far from CLIP's original visual semantics, we set the visual encoder $V(\cdot)$ as a secondary teacher and encourage z_r to remain close to the original visual output:

$$\mathcal{L}_{visual} = \frac{1}{N} \sum_{i=1}^N \left[-\log \frac{\exp(f_{ii}/\tau)}{\sum_{j=1}^N \exp(f_{ij}/\tau)} \right], \quad (20)$$

where τ is a temperature parameter, f_{ij} denotes the cosine similarity between the learned representation z_r^i and the CLIP visual feature F_x^j . The overall objective becomes:

$$\mathcal{L}_{distill} = \lambda_1 \mathcal{L}_{knowledge} + \lambda_2 \mathcal{L}_{visual}, \quad (21)$$

where λ_1 and λ_2 denote the balanced hyperparameters. This dual supervision strategy allows z_r to encode external knowledge from the MLLM while preserving essential visual semantics, resulting in a more robust and semantically aligned distilled representation.

Cross-modal Alignment. After we obtain the distilled representation z_r , we should leverage it to enhance image classification through effective vision-language alignment. Therefore, we leverage OT to achieve fine-grained cross-modal alignment. OT facilitates the alignment of local visual features with diverse prompt features, capturing a broader set of semantic cues. By modeling both

prompt and visual features as discrete distributions, OT computes the distance between these distributions, ensuring a more detailed cross-modal alignment. Specifically, for each class k , we initialize a prompt p_k composed of learnable vectors. We then concatenate z_r with p_k to form the instance-aware knowledge prompt:

$$I_k = \text{concat}(z_r, p_k). \quad (22)$$

This instance-aware knowledge prompt is fed into the text encoder $T(\cdot)$, obtaining the prompt features enriched with instance-aware knowledge:

$$G_k = T(I_k). \quad (23)$$

After we obtain the G_k , we formulate visual features F_x and prompt features G_k as the samplings of two discrete distributions. Then, we apply SSOT [1] to match the G_k with the F_x by minimizing the OT distance. Unlike entropic OT [7], SSOT introduces squared L2-norm regularization to produce sparse transport plans, where many values are zero. This sparsity allows SSOT to clearly identify the most relevant correspondences between visual features and textual features, while filtering out irrelevant ones. For instance, although the distilled representation is designed to be informative, it may still contain redundant information, which may not be relevant for recognizing an object. By ignoring such irrelevant features, SSOT enhances the effectiveness of the alignment process. Specifically, given an image x , we first feed it to the visual encoder $V(\cdot)$ to obtain the visual features $F_x \in \mathbb{R}^{H \times C}$, C is the feature dimension. Simultaneously, for class k , we obtain its instance-aware knowledge prompt features $G_k \in \mathbb{R}^{Z \times C}$ as Eq. (23). To reduce computational cost and avoid introducing extra parameters, we adopt a two-stage optimization strategy to solve the OT problem. In the first stage, we learn the transport matrix P with these support sets F_x , G_k , by minimizing the following SSOT distance to push G_k to F_x as:

$$d_M^\epsilon(k) = \min_{P \in U(\mu, \nu)} \langle P, M \rangle + \frac{\epsilon}{2} \|P\|_2^2, \quad (24)$$

where $M = 1 - F_x G_k^\top$ denotes that we use the cosine distance between F_x and G_k as the cost matrix, $U(\mu, \nu) := \{P \in \mathbb{R}_+^{H \times Z} \mid P \mathbf{1}_H = \mu, P^\top \mathbf{1}_Z = \nu\}$, with μ and ν are the discrete probability vectors that each sum to 1, and ϵ is the coefficient that controls the sparsity of the transport matrix. In our experiments, μ and ν are set as uniform distributions. Given the OT distance between G_k and F_x , we reformulate the prediction probability as:

$$p_{OT}(y = k | x) = \frac{\exp((1 - d_M^\epsilon(k))/\tau)}{\sum_{k'=1}^K \exp((1 - d_M^\epsilon(k'))/\tau)}, \quad (25)$$

τ is a temperature parameter. In the second stage, we fix the transport matrix P and optimize learnable parameters by minimizing the cross-entropy loss:

$$\mathcal{L}_{CE} = -\frac{1}{|X|} \sum_{x \in X} \sum_{k=1}^K y_{x,k} p_{OT}(y = k | x), \quad (26)$$

where $y_{x,k}$ is a one-hot label vector. Through fine-tuning with OT, the learnable vectors are optimized to better integrate with the distilled instance-aware knowledge, enabling the model to dynamically adapt to diverse instances.

Putting Together. The total loss of PLIKD combines the dual-supervised distillation loss and the cross-modal alignment loss, and

is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{distill}} + \lambda_3 \mathcal{L}_{CE}, \quad (27)$$

where λ_3 denotes the balanced hyperparameter. This formulation enables PLIKD to effectively integrate instance-aware knowledge into the prompt learning process, achieving both accurate knowledge transfer and precise cross-modal alignment. Note that we pre-extract the instance-aware knowledge via MLLMs only for the training set and avoid employing MLLMs during both training and inference to reduce the computational burden.

4 Experiments

4.1 Setup

Our approach is mainly evaluated in the following three problem settings: generalization from base to new classes within a dataset (base-to-new generalization), domain generalization [53, 64, 65], and few-shot learning evaluation. Before discussing the results, we provide the details of the experimental setup below.

Datasets and Evaluation Metric. For the few-shot learning and base-to-new generalization settings, following [21, 66, 67], we evaluate performance on 11 publicly available image classification datasets, including ImageNet [9], Caltech101 [13], OxfordPets [40], StanfordCars [31], Flowers102 [39], Food101 [2], FGVC Aircraft [38], SUN397 [58], DTD [6], EuroSAT [16], and UCF101 [46]. For the domain generalization setting, following [36, 67], we train the model on ImageNet and evaluate its performance on four other challenging variants of ImageNet: ImageNetV2 [43], ImageNet-Sketch [52], ImageNet-A [18], and ImageNet-R [17]. Following [29, 36, 66, 67], we consider classification accuracy as the evaluation metric. The details of each dataset are summarized in Appendix C, including the number of classes, the sizes of training and testing sets.

Baselines. We primarily compare our method with the following seven prompt learning baselines: **CoOp** [67], **CoCoOp** [66], **MaPLe** [28], **PromptSRC** [29], **CoPrompt** [44], **ProText** [27], and **PromptKD** [36].

Implementation Details. For the instance-aware knowledge extraction module, we utilize GPT-4o [22] as the knowledge extractor, with the temperature parameter set to 0 to maintain consistency. For the instance-aware knowledge distillation module, we set $\lambda_1=0.7$ and $\lambda_2=0.3$ for Eq. (21). The sparsity coefficient ϵ of SSOT is set to 1.0, and $\lambda_3=0.8$ for Eq. (27). All experiments are repeated three times with different random seeds, and we report the averaged results. More implementation details are provided in the Appendix.

4.2 Comparison with Baselines and Analysis

In this subsection, we compare our PLIKD with the baselines across three problem settings: generalization from base to new classes within a dataset (base-to-new generalization) [32, 45, 48], domain generalization [8, 49], and few-shot learning [35, 37, 50, 63].

Base-to-New Generalization. Table 1 presents the performance comparison on base-to-new generalization. Following [36, 66], we split each dataset into base and new classes, train the model on base classes, and evaluate on both to assess generalization. We report base accuracy (Base), new accuracy (New), and their harmonic mean (H), averaged over three runs. In comparison with previous works, PLIKD shows better performance on 9 of 11 datasets. Notably, on the challenging ImageNet dataset, PLIKD achieves significant

Table 1: Comparison with existing state-of-the-art methods on base-to-new generalization across 11 datasets.

Method	Average			ImageNet			Caltech101			OxfordPets			StanfordCars			Flowers102		
	Base	New	H	Base	New	H	Base	New	H	Base	New	H	Base	New	H	Base	New	H
CoOp	82.69	63.22	71.66	76.47	67.88	71.92	98.00	89.81	93.73	93.67	95.29	94.47	78.12	60.40	68.13	97.60	59.67	74.06
CoCoOp	80.47	71.69	75.83	75.98	70.43	73.10	97.96	93.81	95.84	95.20	97.69	96.43	70.49	73.59	72.01	94.87	71.75	81.71
MaPLe	82.28	75.14	78.55	76.66	70.54	73.47	97.74	94.36	96.02	95.43	97.76	96.58	72.94	74.00	73.47	95.92	72.46	82.56
PromptSRC	84.26	76.10	79.97	77.60	70.73	74.01	98.10	94.03	96.02	95.33	97.30	96.30	78.27	74.97	76.58	98.07	76.50	85.95
CoPrompt	84.00	77.22	80.47	77.67	71.27	74.33	98.27	94.90	96.56	95.67	98.10	96.87	76.97	74.40	75.66	97.27	76.50	85.64
PromptKD	<u>86.96</u>	<u>80.73</u>	<u>83.73</u>	<u>80.83</u>	<u>74.66</u>	<u>77.62</u>	<u>98.91</u>	<u>96.65</u>	<u>97.77</u>	<u>96.30</u>	<u>98.01</u>	<u>97.15</u>	<u>82.80</u>	<u>83.37</u>	<u>83.08</u>	99.42	<u>82.62</u>	<u>90.24</u>
ProText	84.38	74.49	79.13	77.93	71.14	74.38	98.84	94.60	96.67	95.47	97.16	96.31	79.37	74.89	77.06	98.02	76.80	86.12
PLIKD	87.66	82.66	85.09	83.15	77.28	80.11	98.96	97.95	98.45	96.33	98.50	97.40	83.15	85.32	84.22	<u>98.92</u>	84.38	91.07

Method	Food101			FGVCAircraft			SUN397			DTD			EuroSAT			UCF101		
	Base	New	H	Base	New	H	Base	New	H	Base	New	H	Base	New	H	Base	New	H
CoOp	88.33	82.26	85.19	40.44	22.30	28.75	80.60	65.89	72.51	79.44	41.18	54.24	92.19	54.74	68.69	84.69	56.05	67.46
CoCoOp	90.70	91.29	90.99	33.41	23.71	27.74	79.74	76.86	78.27	77.01	56.00	64.85	87.49	60.04	71.21	82.33	73.45	77.64
MaPLe	90.71	92.05	91.38	37.44	35.61	36.50	80.82	78.70	79.75	80.36	59.18	68.16	94.07	73.23	82.35	83.00	78.66	80.77
PromptSRC	90.67	91.53	91.10	42.73	37.87	40.15	82.67	78.47	80.52	83.37	62.97	71.75	92.90	73.90	82.32	87.10	78.80	82.74
CoPrompt	90.73	92.07	91.40	40.20	39.33	39.76	82.63	80.03	81.31	83.13	64.73	72.79	94.60	78.57	85.84	86.90	79.57	83.07
PromptKD	<u>92.43</u>	<u>93.68</u>	<u>93.05</u>	<u>49.12</u>	<u>41.81</u>	<u>45.17</u>	<u>83.69</u>	<u>81.54</u>	<u>82.60</u>	<u>85.84</u>	<u>71.37</u>	<u>77.94</u>	97.54	<u>82.08</u>	<u>89.14</u>	<u>89.71</u>	<u>82.27</u>	<u>85.83</u>
ProText	90.75	91.31	91.03	43.19	36.29	39.44	81.36	72.35	76.59	83.24	62.90	71.65	93.48	64.15	76.09	86.53	77.82	81.94
PLIKD	93.40	95.76	94.57	50.07	43.39	46.49	84.82	82.93	83.86	87.19	73.83	79.96	<u>96.94</u>	85.82	91.04	91.33	84.15	87.59

Table 2: Experimental results on domain generalization.

	Source	Target			
	ImageNet	-V2	-Sketch	-A	-R
CoOp	71.51	64.20	47.99	49.71	75.21
CoCoOp	71.02	64.07	48.75	50.63	76.18
MaPLe	70.72	64.07	49.15	50.90	76.98
PromptSRC	71.27	64.35	49.55	50.90	77.80
CoPrompt	70.80	64.25	49.43	50.50	77.51
PromptKD	<u>72.29</u>	<u>66.85</u>	<u>50.16</u>	<u>51.77</u>	<u>78.06</u>
ProText	71.31	64.52	49.65	50.96	<u>79.11</u>
PLIKD	74.39	69.20	52.93	53.85	80.90

improvements, demonstrating its ability to generalize effectively to new classes. We attribute this success to the integration of distilled instance-aware knowledge with learnable prompts, enabling the model to dynamically adapt to each instance.

Domain Generalization. For the domain generalization setting, we fine-tune the model on ImageNet under a 16-shot setting and evaluate its performance on four ImageNet variants: ImageNetV2 [43], ImageNet-Sketch [52], ImageNet-A [18], and ImageNet-R [17], and we report the classification accuracy. Table 2 summarizes the performance comparison of PLIKD with baseline methods. In comparison with previous works, PLIKD clearly outperforms all baselines on all target datasets. Notably, compared to the most related work CoCoOp, PLIKD achieves higher accuracy margins on ImageNetV2 (+5.13%), ImageNet-Sketch (+4.18%), ImageNet-A (+3.22%), and ImageNet-R (+4.72%). These results confirm that PLIKD is more domain-generalizable than previous methods, highlighting its ability to maintain strong performance under significant domain shifts by effectively learning robust visual representations.

Few-Shot Learning. To evaluate the few-shot learning performance of our PLIKD, we conduct experiments on 11 image recognition datasets following [66, 67] and compare it with the baselines. Detailed results and analysis can be found in the Appendix.

4.3 Ablation Study

Instance-aware Knowledge. To assess the impact of multidimensional instance-aware knowledge, we conduct an ablation study by removing each of the three components: visual features, functional attributes, and contextual information. Fig. 3(a) shows the average performance across 11 datasets, and Fig. 3(b) presents the results on the ImageNet. It can be observed that excluding any component leads to a performance drop, with the largest drop observed when removing the visual dimension. The results suggest that each component contributes significantly to the model’s performance, with the visual dimension as the most informative single modality, and the inclusion of all three providing the most benefit for both base and new classes. This demonstrates the importance of integrating multidimensional instance-aware knowledge in our method.

Dual-supervised Distillation. We compare PLIKD with dual-supervised distillation to a variant with only MLLM supervision in Fig. 4(a) and Fig. 4(b). PLIKD achieves consistently better base and new accuracies across 1, 2, 4, 8, and 16-shot settings (averaged over 11 datasets). The results suggest that dual-supervised distillation improves both accuracy and generalization by integrating instance-aware knowledge from MLLMs while preserving CLIP’s visual semantics. The MLLM supervision facilitates external knowledge integration, while CLIP supervision prevents overfitting to MLLM signals and maintains alignment with the original visual semantics, thereby enhancing performance on both base and novel categories. **Smooth and Sparse Optimal Transport.** We conduct an ablation study by comparing the performance of our method with SSOT against a variant with another OT formulation, i.e., entropic OT [7].

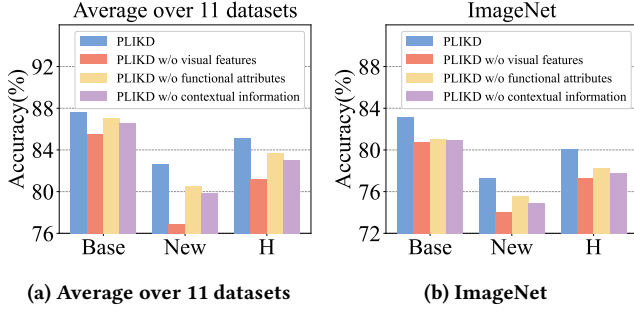


Figure 3: Ablation study on instance-aware knowledge.

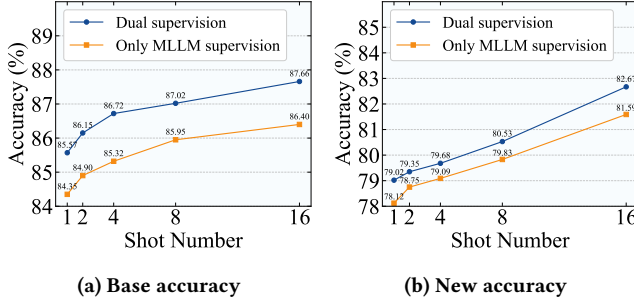


Figure 4: Ablation study on dual-supervised distillation.

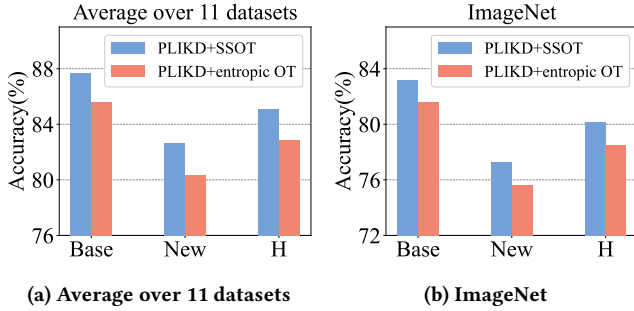


Figure 5: Ablation study on two OT formulations.

We evaluate the base-to-new generalization results averaged over the 11 datasets, along with the results on the challenging ImageNet dataset, as illustrated in Fig. 5. It can be observed that our method with SSOT significantly outperforms the entropic OT variant in both base and new class accuracy. This suggests that SSOT facilitates more accurate feature matching, leading to better generalization across seen and unseen classes. These findings underscore its crucial role in fine-grained cross-modal alignment within our framework.

4.4 Analysis

Sparsity Coefficient in SSOT. To investigate the impact of the sparsity coefficient ϵ in SSOT, we conduct ablation experiments by varying ϵ in the range $\{0.5, 1.0, 2.0, 3.0\}$. In the experiments, we tune this hyperparameter on the challenging ImageNet dataset [9] and apply it to other datasets. The performance peaks when $\epsilon = 1.0$, which justifies our default setting in the main experiments. Figure 6 shows the detailed results.

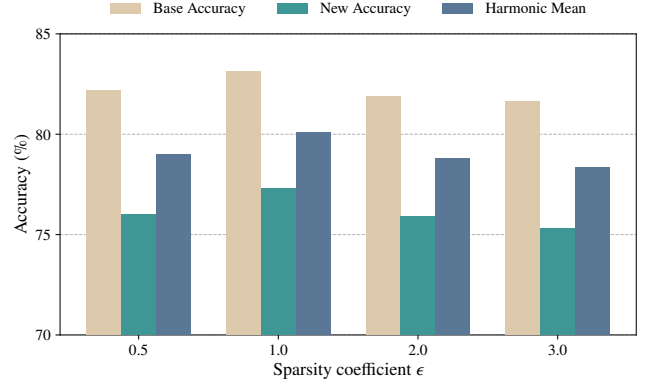


Figure 6: Effect of the sparsity coefficient in SSOT.

Table 3: Comparison of computation time cost.

Method	CoCoOp	ProText	PLIKD
Training Time (s)	1.155	1.182	1.157
Inference Time (images/s)	706.2	653	705

Parameter Sensitivity. We study the effects of the loss function hyperparameters (λ_1 , λ_2 , and λ_3) on model performance. The corresponding results and analysis are provided in the Appendix to better illustrate their impact.

Computation Time Cost. As shown in Table 3, we compare the training and inference efficiency of CoCoOp, ProText, and our PLIKD. We report the one-epoch training time under the 1-shot setting on the Food101 dataset. PLIKD achieves a comparable training time to CoCoOp and faster than ProText.

5 Conclusion

In this paper, we propose PLIKD, a novel prompt learning method that integrates instance-aware knowledge distillation for Web-scale semantic image classification with unseen classes and distribution shifts. PLIKD involves an instance-aware knowledge extraction module and an instance-aware knowledge distillation module. In the instance-aware knowledge extraction module, we leverage MLLMs with CoT prompting for knowledge extraction. In the instance-aware knowledge distillation module, we employ a dual-teacher strategy to transform the extracted instance-aware knowledge into a robust distilled representation, and apply SSOT for more accurate matching of prompt and image features. We also conduct extensive experiments to demonstrate the effectiveness of our proposed PLIKD.

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Appendix

A Related Work

Vision-Language Models. VLMs have achieved remarkable success in open-world visual concept learning [10, 34]. These models [24, 42, 62] are pretrained on large-scale image-text datasets, which enable them to learn semantic correspondences between visual and linguistic modalities. For instance, CLIP [42] is trained with 400 million image-text pairs using contrastive learning [5, 30], while ALIGN [24] leverages 1.8 billion noisy image-text pairs. These models exhibit strong open-vocabulary understanding capabilities [56, 57] and can naturally adapt to a variety of downstream classification tasks [33, 58].

Prompt Learning for Image Classification. Prompt learning [25, 28, 41, 59] has emerged as a more efficient alternative to fine-tuning CLIP towards image classification tasks. CoOp [67] is the first to incorporate learnable prompts into vision-language models for few-shot classification. CoCoOp [66] further enhances performance on new categories by incorporating instance-level image information. MaPLe [28] adopts multi-modal prompt learning for both vision and language branches to improve alignment between the vision and language features. PromptSRC [29] proposes a regularization framework to prevent prompt learning from overfitting. CoPrompt [44] incorporates consistency constraints with prompt learning to reduce the overfitting problem and improve generalization. PromptKD [36] performs domain-specific prompt-based knowledge distillation [12] for CLIP using unlabeled domain data. ProText [27] proposes a new direction to adapt CLIP by learning generalized prompts with text-only supervision.

B Modeling for PLIKD

Knowledge Extraction from MLLMs. To ensure the MLLMs generate outputs that align with our requirements, we incorporate explicit examples within Question 1 to illustrate the desired response format and content.

Question 1: *Based on the given class name, describe its typical visual features, functional attributes, and contextual information. Be concise and ensure accuracy.*

This step can be formulated as follows:

$$k = \text{MLLMs}(\text{Question1, class name}). \quad (28)$$

Below is a representative example employed in Question 1:

Example: *Class: airplanes. Visual Features: Wings, fuselage, engines, tail. Functional Attributes: Used for air transportation. Contextual Information: Airports, blue sky.*

Through experimentation, we found that including just a single example is sufficient to steer the model toward generating high-quality and consistent outputs. Therefore, in all of our experiments, we use a single example throughout all experiments. As a result, we adopt the following template as the default input to the model:

Question 1: *Based on the given class name, describe its typical visual features, functional attributes, and contextual information. Be concise and ensure accuracy. Example: Class: airplanes. Visual*

Features: Wings, fuselage, engines, tail. Functional Attributes: Used for air transportation. Contextual Information: Airports, blue sky.

Algorithm 2 The training process of PLIKD

- 1: **Input:** Image data: x ; corresponding instance-aware knowledge q_m ; pretrained visual encoder $V(\cdot)$ and textual encoder $T(\cdot)$; the learnable visual prompt l_v and a learnable feature projector $g(\cdot)$; the learnable textual prompt p_k ; maximum number of iterations in first and second stage $O_{\text{fir}}, O_{\text{sec}}$, hyperparameters: $\lambda_1, \lambda_2, \lambda_3$.
 - 2: Obtain textual features of q_m : $t_f = T(q_m)$;
 - 3: Obtain the instance representation: $z_r = g(V(x; l_v))$;
 - 4: Obtain the visual features for the input image x : $F_x = V(x)$;
 - 5: Obtain the MLLM supervision loss: $\mathcal{L}_{\text{knowledge}}$;
 - 6: Obtain the CLIP supervision loss: $\mathcal{L}_{\text{visual}}$;
 - 7: Obtain the total distillation loss: $\mathcal{L}_{\text{distill}} = \lambda_1 \mathcal{L}_{\text{knowledge}} + \lambda_2 \mathcal{L}_{\text{visual}}$;
 - 8: **for** $t_{\text{out}} = 1, 2, \dots, O_{\text{sec}}$ **do**
 - 9: Obtain a visual feature set $F \in \mathbb{R}^{H \times C}$ with the visual encoder $V(\cdot)$;
 - 10: Generate prompt feature set $G_k \in \mathbb{R}^{Z \times C}$ of each class k with the textual encoder $T(\cdot)$;
 - 11: Calculate the cost matrix $C_k = 1 - F^T G_k \in \mathbb{R}^{H \times Z}$ of each class;
 - 12: **for** $t_{\text{in}} = 1, 2, \dots, O_{\text{fir}}$ **do**
 - 13: Calculate the OT distance until coverage;
 - 14: **end for**
 - 15: Obtain optimal transport plan as Alg1;
 - 16: Calculate the OT distance;
 - 17: Calculate the classification probability $p_{\text{OT}}(y = k|x)$ with the OT distance;
 - 18: **end for**
 - 19: Obtain the cross-entropy loss $\mathcal{L}_{\text{CE}}$;
 - 20: Obtain the total loss: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{distill}} + \lambda_3 \mathcal{L}_{\text{CE}}$;
 - 21: Minimize loss function $\mathcal{L}_{\text{total}}$;
 - 22: Update $l_v, g(\cdot)$ and p_k ;
 - 23: **Return:** The optimal $l_v, g(\cdot)$ and p_k .
-

Detailed algorithm of the training process. The training process of PLIKD is provided in Algorithm 2.

Table 4: Dataset statistics for Web-scale evaluation.

Dataset	Classes	Training size	Testing size
Caltech101	100	4128	2465
DTD	47	2820	1692
FGVCAircraft	100	3334	3333
Food101	101	50500	30300
EuroSAT	10	13500	8100
Flowers102	102	4093	2463
OxfordPets	37	2944	3669
StanfordCars	196	6509	8041
SUN397	397	15880	19850
UCF101	101	7639	3783
ImageNet	1000	1.28M	50000

C Experiments

Datasets and Evaluation Metric. For the few-shot learning and base-to-new generalization settings, following [21, 66, 67], we evaluate performance on 11 publicly available image classification datasets, including ImageNet [9], Caltech101 [13], OxfordPets [40], StanfordCars [31], Flowers102 [39], Food101 [2], FGVCAircraft [38], SUN397 [58], DTD [6], EuroSAT [16], and UCF101 [46]. For the domain generalization setting, following [36, 67], we train the model on ImageNet and evaluate its performance on four other challenging variants of ImageNet: ImageNetV2 [43], ImageNet-Sketch [52], ImageNet-A [18], and ImageNet-R [17]. Following [29, 36, 66, 67], we consider classification accuracy as the evaluation metric. The details of each dataset are shown in Table 4, including the number of classes, the sizes of training and testing sets. **Baselines.** We primarily compare our method with the following seven prompt learning baselines: **CoOp** [67], **CoCoOp** [66], **MaPLe** [28], **PromptSRC** [29], **CoPrompt** [44], **ProText** [27], and **PromptKD** [36]. CoOp [67] is the first to incorporate learnable prompts into VLMs for few-shot classification. CoCoOp [66] further enhances performance on new categories by incorporating instance-level image information. MaPLe [28] adopts multi-modal prompt learning for both vision and language branches to improve alignment between the vision and language features. PromptSRC [29] proposes a regularization framework to prevent prompt learning from overfitting. CoPrompt [44] incorporates consistency constraints with prompt learning to reduce the overfitting problem and improve generalization. PromptKD [36] performs domain-specific prompt-based knowledge distillation [60] for CLIP using unlabeled domain data. ProText [27] proposes a new direction to adapt CLIP by learning generalized prompts with text-only supervision.

Implementation Details. We utilize GPT-4o [22] as the knowledge extractor, with the temperature parameter set to 0 to maintain consistency. We set $\lambda_1 = 0.7$ and $\lambda_2 = 0.3$. We initialize the learnable prompts following CoOp [67], adopting the following configurations: the “end” token position, random initialization, 16 context tokens, and the ViT-B/16 backbone. We set the sparsity coefficient ϵ of SSOT to 1.0. Finally, we set $\lambda_3 = 0.8$. To ensure fair comparison, all baselines use the same backbone as our method. All experiments are repeated three times with different random seeds, and we report the average results. The original CoOp method has multiple versions that differ in class token positions and parameter initialization strategies. Since the performance gap among these versions is limited, we select a specific version as our baseline. This version uses the “end” token position, the parameter initialization strategy is “random”, the length of learnable context tokens is set as 16, and the ViT-B/16 backbone. All the code of our method is based on CoOp, which adopted the SGD optimizer with a 0.002 initial learning rate, CosineAnnealingLR schedule, and a warmup trick with a 1e-5 learning rate. We also followed the epoch strategy to train more epochs for more shots. For small datasets such as FGVCAircraft, Flowers102, and StanfordCars, the batch size is set as 32, while for the larger dataset such as ImageNet and SUN397, the batch size is set as 128. We set the maximum iteration number of the first stage as 100. We initialize all values in the vector v and μ as $1/N$ and $1/M$ respectively. Since CLIP performs poorly on long

Table 5: Comparison with baseline methods on few-shot learning. AVG represents the average performance across datasets.

Method	ImageNet	Caltech101	OxfordPets	Cars	Flowers102	Food101	Aircraft	SUN397	DTD	EuroSAT	UCF101	AVG
CoOp	61.91	91.99	87.02	73.60	94.49	74.48	31.43	68.36	62.51	83.69	76.90	73.33
CoCoOp	60.47	91.52	87.15	70.19	93.27	77.02	31.28	68.07	61.89	83.15	76.59	72.78
PromptSRC	62.86	92.03	87.17	73.15	94.80	76.97	31.62	70.83	66.08	84.02	77.40	74.27
PromptKD	<u>65.26</u>	<u>92.31</u>	<u>87.38</u>	<u>74.89</u>	<u>95.17</u>	<u>77.93</u>	<u>33.83</u>	<u>72.06</u>	<u>68.32</u>	<u>85.33</u>	<u>77.85</u>	<u>75.49</u>
ProText	65.01	92.24	87.21	73.80	94.76	77.09	32.49	71.96	67.60	85.23	77.26	75.05
PLIKD	67.39	92.58	87.62	76.52	95.19	79.02	35.76	74.19	69.93	85.86	78.39	76.59

Table 7: Effect of different λ_3 on model performance.

λ_3	Base	New	HM
1.0	82.70	77.06	79.78
0.9	82.95	77.13	79.93
0.8	83.15	77.28	80.11
0.7	82.86	77.08	79.87
0.6	82.63	76.92	79.67
0.5	82.59	76.83	79.61
0.4	82.50	76.35	79.31
0.3	82.46	76.30	79.26
0.2	82.31	76.23	79.15

text, we retain only the first 64 tokens of each MLLM output to comply with its input length limit. All models are implemented in PyTorch 1.7.1 and trained on four NVIDIA A100 GPUs.

C.1 Comparison with Baselines and Analysis

Few-Shot Learning. To evaluate the few-shot learning performance of our method, we conduct experiments on 11 image recognition datasets following the [4, 67]. Specifically, we train the model with 16-shot settings and evaluate it on the original test sets. We report the classification accuracy to evaluate the model’s performance. The results and comparisons are shown in Table 5. For brevity, we abbreviate StanfordCars and FGVC Aircraft as Cars and Aircraft, respectively. It is clear that PLIKD outperforms all baselines, including PromptKD, which incorporates knowledge distillation. Taking

the average (AVG) accuracy as an example, PLIKD achieves performance improvements of 3.26% over CoOp, 3.81% over CoCoOp, 2.32% over PromptSRC, 2.65% over ProText, and 1.10% over PromptKD respectively. We attribute this improvement to the fact that, unlike existing prompt learning methods optimized solely for class-level information, PLIKD is instance-aware, leveraging both class-specific and instance-specific knowledge. All these performance comparisons provide experimental evidence that PLIKD effectively addresses the limitations of existing prompt learning methods and achieves enhanced generalization in few-shot learning scenarios.

Table 6: Effect of λ_1 and λ_2 on model performance.

λ_1	λ_2	Base	New	HM
0.9	0.1	82.05	75.83	78.82
0.8	0.2	82.20	76.34	79.16
0.7	0.3	83.15	77.28	80.11
0.6	0.4	82.63	76.50	79.45
0.5	0.5	82.12	76.30	79.10
0.4	0.6	81.90	76.05	78.87
0.3	0.7	81.77	75.63	78.58
0.2	0.8	81.52	75.39	78.34
0.1	0.9	81.15	74.89	77.89

Parameter Sensitivity. We study the effects of the loss function hyperparameters (λ_1 , λ_2 , and λ_3) on model performance. We vary these hyperparameters on ImageNet and report base accuracy (Base), new accuracy (New), and their harmonic mean (HM), averaged over three runs. Table 6 and Table 7 show the detailed results.